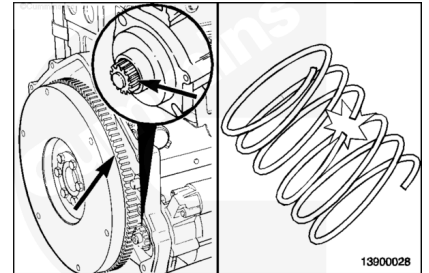


Trainings ▾

Initial Check

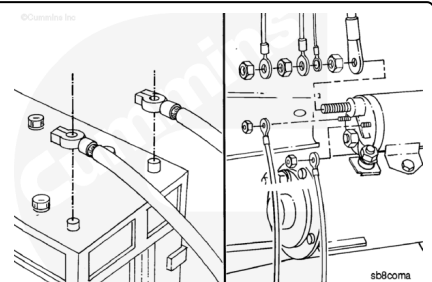
If the starter solenoid is making a sound but the engine is **not** rotating, remove the starter motor and check for broken ring gear teeth or a broken starter motor spring.



Remove

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing the batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

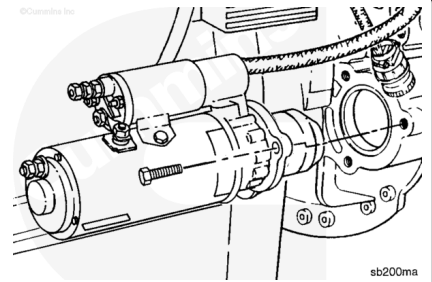
Remove the electrical connections from the batteries, negative (-) cable first.

Remove the electrical connections from the starter motor.

Identify each wire with a tag indicating location on starter motor.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



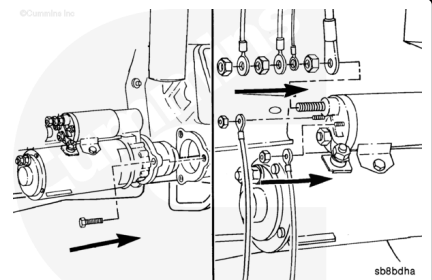
Remove the capscrews from the flywheel housing.

Remove the starter motor.

Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



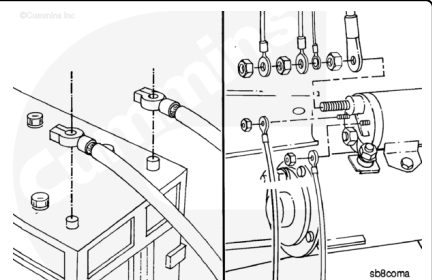
Install the starter motor.

Install and tighten the mounting capscrews.

Torque Value: 43 n•m [32 ft-lb]

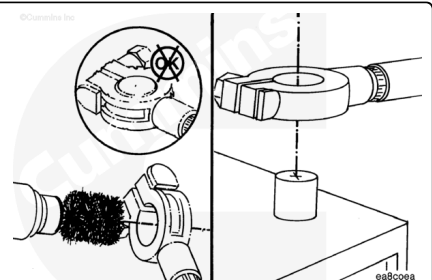
Note : Use the location tags to help identify where each wire connection goes.

Install the starter motor electrical connections.



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⚠ WARNING ⚠

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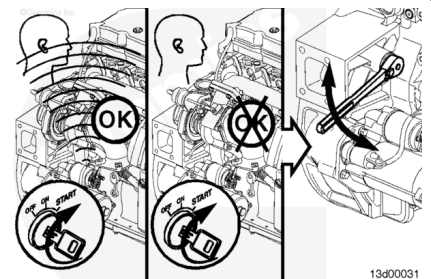
Clean the inside of the battery terminal with a brush before connecting to battery.

Install battery cables, connecting the negative (-) cable last.

Rotation Check

If the starter motor solenoid makes a sound, turn the keyswitch to the OFF position, and attempt to bar the crankshaft in both directions.

Bar the engine using the barring tool, Part Number 3824591.



Bar the crankshaft as follows:

1. Direction of engine rotation
2. Direction opposite engine rotation
3. Direction of engine rotation.

If the crankshaft will bar over, attempt to start the engine. If the starter motor cranks the engine, check the starter motor pinion gear and flywheel ring gear for damage.

